

Seamless:

Multilingual Expressive and Streaming Speech Translation

Seamless Communication, Loïc Barrault*, Yu-An Chung*, Mariano Coria Meglioli*, David Dale*, Ning Dong*, Mark Duppenthaler*, Paul-Ambroise Duquenne*[‡], Brian Ellis*, Hady Elsahar*, Justin Haaheim*, John Hoffman*, Min-Jae Hwang*, Hirofumi Inaguma*, Christopher Klaiber*, Ilia Kulikov*, Pengwei Li*, Daniel Licht*, Jean Maillard*, Ruslan Mavlyutov*, Alice Rakotoarison*, Kaushik Ram Sadagopan*, Abinеш Ramakrishnan*, Tuan Tran*, Guillaume Wenzek*, Yilin Yang*, Ethan Ye*, Ivan Evtimov, Pierre Fernandez, Cynthia Gao, Prangthip Hansanti, Elahe Kalbassi, Amanda Kallet, Artyom Kozhevnikov, Gabriel Mejia Gonzalez, Robin San Roman, Christophe Touret, Corinne Wong, Carleigh Wood, Bokai Yu, Pierre Andrews[†], Can Balioglu[†], Peng-Jen Chen[†], Marta R. Costa-jussà[†], Maha Elbayad[†], Hongyu Gong[†], Francisco Guzmán[†], Kevin Heffernan[†], Somya Jain[†], Justine Kao[†], Ann Lee[†], Xutai Ma[†], Alex Mourachko[†], Benjamin Peloquin[†], Juan Pino[†], Sravya Popuri[†], Christophe Ropers[†], Safiyyah Saleem[†], Holger Schwenk[†], Anna Sun[†], Paden Tomasello[†], Changhan Wang[†], Jeff Wang[†], Skyler Wang^{†§}, Mary Williamson[†]

FAIR at Meta, [‡]INRIA, [§]UC Berkeley

*Equal contribution, alphabetical order

[†]Research and engineering leadership—equal contribution, alphabetical order.

Recent advancements in automatic speech translation have dramatically expanded language coverage, improved multimodal capabilities, and enabled a wide range of tasks and functionalities. That said, large-scale automatic speech translation systems today lack key features that help machine-mediated communication feel seamless when compared to human-to-human dialogue. In this work, we introduce a family of models that enable end-to-end *expressive* and multilingual translations in a *streaming* fashion. First, we contribute an improved version of the massively multilingual and multimodal SEAMLESSM4T model—SEAMLESSM4T v2. This newer model, incorporating an updated UNITY2 framework, was trained on more low-resource language data. The expanded version of SEAMLESSALIGN adds 114,800 hours of automatically aligned data for a total of 76 languages. SEAMLESSM4T v2 provides the foundation on which our two newest models, SEAMLESSEXPRESSIVE and SEAMLESSSTREAMING, are initiated. SEAMLESSEXPRESSIVE enables translation that preserves vocal styles and prosody. Compared to previous efforts in expressive speech research, our work addresses certain underexplored aspects of prosody, such as speech rate and pauses, while also preserving the style of one’s voice. As for SEAMLESSSTREAMING, our model leverages the Efficient Monotonic Multihead Attention (EMMA) mechanism to generate low-latency target translations without waiting for complete source utterances. As the first of its kind, SEAMLESSSTREAMING enables simultaneous speech-to-speech/text translation for multiple source and target languages. To understand the performance of these models, we combined novel and modified versions of existing automatic metrics to evaluate prosody, latency, and robustness. For human evaluations, we adapted existing protocols tailored for measuring the most relevant attributes in the preservation of meaning, naturalness, and expressivity. To ensure that our models can be used safely and responsibly, we implemented the first known red-teaming effort for multimodal machine translation, a system for the detection and mitigation of added toxicity, a systematic evaluation of gender bias, and an inaudible localized watermarking mechanism designed to dampen the impact of deepfakes. Consequently, we bring major components from SEAMLESSEXPRESSIVE and SEAMLESSSTREAMING together to form SEAMLESS, the first publicly available system that unlocks expressive cross-lingual communication in real-time. In sum, SEAMLESS gives us a pivotal look at the technical foundation needed to turn the Universal Speech Translator from a science fiction concept into a real-world technology. Finally, contributions in this work—including models, code, and a watermark detector—are publicly released and accessible at the link below.

Date: November 30, 2023

Correspondence: Xutai Ma at xutaima@meta.com

Code: https://github.com/facebookresearch/seamless_communication



Contents

1	Introduction	3
2	Beyond Words: Expressive and Streaming Speech-to-Speech Translation	5
2.1	Towards Naturalistic Speech-to-Speech Translation	5
2.2	Expressive and Streaming S2ST Today	7
2.3	Overview of Model Capabilities & Languages	9
3	SEAMLESSM4T v2	12
3.1	Data for Speech Translation	13
3.2	Pre-Training	16
3.3	Predicting Units with UNITY2	17
3.4	S2ST Training Setup.	20
3.5	Results and Discussion	21
4	SEAMLESSEXPRESSIVE	25
4.1	Expressive Speech-to-Speech Translation Data	25
4.2	Expressive Modeling	31
4.3	Experimental Setup	36
4.4	Results and Discussion	38
4.5	Ablation Study	41
5	SEAMLESSSTREAMING	45
5.1	Efficient Monotonic Multihead Attention (EMMA)	45
5.2	Experimental Setup	47
5.3	Results and Discussion	49
6	SEAMLESS	52
6.1	Architecture	53
6.2	Results and Discussion	53
7	Automatic and Human Evaluation	56
7.1	Automatic Expressivity Metrics	56
7.2	Robustness Automatic Evaluation	58
7.3	Human Evaluation	59
8	Responsible AI	66
8.1	Red Teaming	67
8.2	Toxicity	70
8.3	Gender Bias	77
8.4	Localized Watermarking	79
9	Social Impact & Conclusion	83
9.1	Naturalistic Translations & Experiential Futures	83
9.2	Ethical Considerations & Future Work	84
	Appendices	101

1. Introduction

German literary critic Friedrich Schlegel once said, “What is lost in the good or excellent translation is precisely the best.” When applied to speech, this sentiment implies that even when a translation accurately renders the semantic meaning of an utterance, certain defining elements of speech may be lost in the process (Schuller et al., 2013).

While the specific constituents of what Schlegel deemed *the best* are open for interpretation, the speech translation research community has long homed in on two components: the *indexical* (i.e., components marking the characteristics of a person) and *pragmatical* (i.e., the way communication works in social situations) components of speech that make human communication what it is. For speech to be natural, it relies on the indexical or revelatory nature of the human voice (Costello, 2000). A speech translation system that incorporates features that help a listener make inferences about a speaker’s personhood bolsters the naturalness of a machine-mediated interaction (Waytz et al., 2014). Preserving vocal style also involves capturing the prosodic elements of speech (e.g., pitch, stress, rhythm), which are key in facilitating the expression of meaning, emotions, and intent (Aguero et al., 2006; Anumanchipalli et al., 2012). Next, human speech and translation are sensitive to pragmatic nuances such as turn-taking and timing controls (Cokely, 1986; Levinson, 2016). Picture how human simultaneous interpreters work: they find just the right balance between low-latency *and* accurate translations. Waiting too long stifles the flow of communication, while going too fast compromises the overall quality of a translation.

Existing research efforts aimed at preserving these intrinsically human features in translation have led to the independent development of expressive and streaming speech-to-speech translation (S2ST) systems. On the expressive front, recent advances in text-to-speech synthesis have integrated voice style transfer via speech language model (Wang et al., 2023a; Kharitonov et al., 2022), flow matching (Le et al., 2023) and diffusion model (Shen et al., 2023). These approaches subsequently inspired S2ST models designed to preserve the source speech’s vocal style and style qualities with a cascaded architecture. Despite these advances, an open, comprehensive S2ST system capturing semantic translation, rhythm, pauses, and sentence-level preservation of the style of one’s voice has yet to be realized. Streaming wise, recent efforts have explored how different simultaneous translation policies (e.g., rule-based or learnable policies) could be deployed to produce systems that strike a balance between low latency and high-quality translations (Ma et al., 2019a; Arivazhagan et al., 2019; Ma et al., 2020c). That said, existing research investments in streaming have homed in on speech-to-text translation (S2TT), and the few that are S2ST compatible are limited in language coverage. Moreover, most streaming translation systems focus on bilingual communication, limiting their utility in contexts where a group of speakers converse in multiple different languages.

To advance research in multilingual expressive and streaming speech translation, we introduce **SeamlessM4T v2**, **SeamlessExpressive**, and **SeamlessStreaming**. SEAMLESSM4T v2 is the foundational multilingual and multimodal model on which the latter two models are initialized. As an improved version of SEAMLESSM4T, SEAMLESSM4T v2 delivers state-of-the-art semantic accuracy across different speech and text translation tasks while supporting nearly 100 languages as input speech or text. This new version features multitask-UNITY2 with its non-auto-regressive unit decoder and hierarchical upsampling, making predicting units much more data-efficient. The new w2v-BERT 2.0 speech encoder of SEAMLESSM4T v2 was pre-trained on 4.5M hours of unlabeled audio data, and the multitask model was finetuned with more supervision from automatically aligned pairs to boost SEAMLESSM4T v2’s performance on low-resource languages. Built using commissioned and publicly available datasets, SEAMLESSEXPRESSIVE enables translation that preserves vocal style and prosody (e.g., rhythm and tone). The model supports translations from and into English in five languages. To our knowledge, SEAMLESSEXPRESSIVE is the first model to enable expressive S2ST from *and* into English and supports underexplored aspects of prosody such as speech rate and pauses. Our SEAMLESSSTREAMING model leverages the Efficient Monotonic Multihead Attention (EMMA) (Ma et al., 2023) mechanism to generate low-latency target translations without waiting for complete source utterances. As the first of its kind to provide many-to-many translations in a simultaneous manner, SEAMLESSSTREAMING supports the same language coverage as the scale of SEAMLESSM4T v2 in ASR, S2TT, and S2ST tasks.

To comprehensively evaluate our systems, we combined existing and newly developed metrics (Section 9.2). For expressivity, we developed two new automatic metrics that measure prosody—AUTOPCP and a rhythm

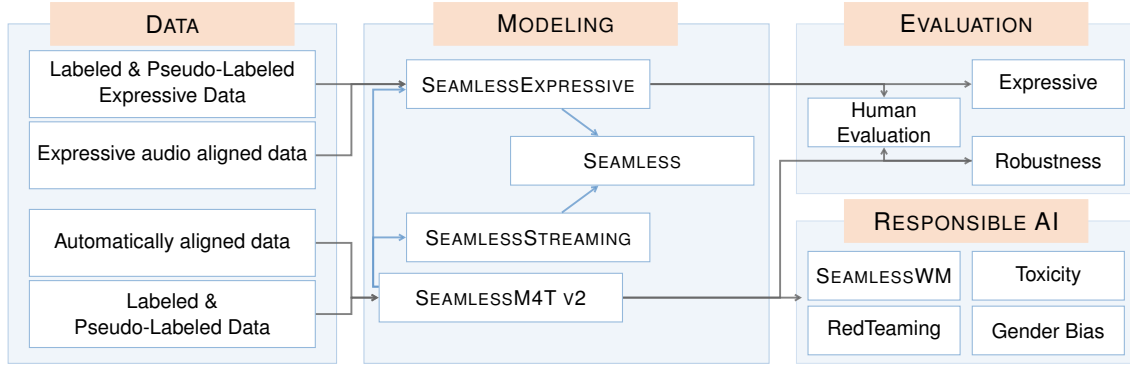


Figure 1 - An overview of the technical components of SEAMLESS and how they fit together.

evaluation toolkit. For human evaluation, we used Cross-lingual Semantic Textual Similarity (XSTS) (Licht et al., 2022) to measure semantics, Mean Opinion Score (MOS) to measure the speech quality of all of our models, and a modified version of the Prosodic Consistency Protocol (PCP) (Huang et al., 2023) to measure the extent to which the expressive qualities in source and target audio are matched. For latency, we used Ending Offset (see Section 5.2.3) for speech output (i.e., the time between when a person finishes speaking and the last translated speech being generated) and an adapted version of Average Lagging (Ma et al., 2019a, 2020b) (i.e., a metric that quantifies the degree to which a listener is out of sync with a speaker with regards to the number of seconds in the source speech) and Length-Adaptive Average Lagging (Papi et al., 2022) for text output. Moreover, we used well-known metrics such as BLEU, chrF, and BLASER 2.0 to measure translation quality automatically. Lastly, we tested for robustness towards noise and vocal style variations.

To ensure that our models are built safely and ethically, we took a four-pronged approach to Responsible AI by implementing 1) the first known red-teaming effort for machine translation, 2) added-toxicity detection and mitigation, 3) a systematic evaluation of gender bias, and 4) an inaudible, localized watermarking mechanism named SEAMLESSWM. We also introduce the new concept of a *metric card* (Section 9.2) that compiles details of our evaluation and Responsible AI metrics.

Combining these building blocks, our unified model **Seamless** (comprising of SEAMLESSEXPRESSIVE and SEAMLESSSTREAMING) marks the first publicly available system that unlocks expressive cross-lingual communication in real-time (see Figure 1). Crucially, SEAMLESS gives us a pivotal look at the technical foundation needed to transform the Universal Speech Translator from a science fiction concept to a real-world technology. To spur further research into related domains and make our work available to the various communities that could benefit from our effort, we publicly release the following at https://github.com/facebookresearch/seamless_communication:

- Models & code: SEAMLESSM4T v2, SEAMLESSEXPRESSIVE, SEAMLESSSTREAMING, and SEAMLESS models
- Automatically aligned data models, code, and metadata: SEAMLESSALIGN data and SONAR speech encoders
- Evaluation tools: AUTOPCP, rhythm evaluation toolkit, and a multilingual alignment extraction toolkit based on the Unity2 aligner
- Responsible AI tools: SEAMLESSWM detector

The rest of the article is structured as follows: Section 2 contextualizes the sociotechnical need for expressive and streaming speech translation via an interview study with users who experience language barriers in their day-to-day lives. Then, it outlines existing technical efforts that tackle this issue, followed by a list of tasks and languages our models support. Section 3 details the various improvements made to SEAMLESSM4T to create SEAMLESSM4T v2. Section 4 and Section 5 detail the data and modeling techniques devised to train models that supports both expressive and streaming multilingual translations. Section 6 reports how we bring SEAMLESSEXPRESSIVE and SEAMLESSSTREAMING together to form SEAMLESS. Subsequently,

Section 7 documents the automatic and human evaluation of our translation outputs, and the robustness of our models in various settings. Section 8 homes in on our Responsible AI effort, where we provide details on our red-teaming, added toxicity detection and mitigation, gender bias evaluation, and watermarking efforts. Finally, we conclude in Section 9, where we discuss the social impact of our work and offer a forward-looking perspective on how SEAMLESS could spearhead the transformation of multilingual communication in the near future.

2. Beyond Words: Expressive and Streaming Speech-to-Speech Translation

In this section, we discuss the sociotechnical need for and the current technical landscape behind developing systems that facilitate expressive and streaming speech-to-speech translation. Then, we outline our contributions by summarizing the capabilities and language coverage for each of our models.

2.1 Towards Naturalistic Speech-to-Speech Translation

For long, investments in natural language processing (NLP) and machine translation research have coalesced around the text modality (NLLB Team et al., 2022; Seamless Communication et al., 2023). While this has given rise to systems that help us translate books, webpages, and text messages, speech translation has lagged behind in terms of language coverage and performance. As a denser modality, the very paralinguistic features (e.g., prosody, tone, timing controls, etc.) that make speech challenging from a computational perspective are also why S2ST systems are filled with promises (Kraut et al., 1992; Nakamura, 2009). The consummate system, which resembles the fictional Universal Speech Translator in *Star Trek*, would seamlessly offer expressive and real-time translation without excessive tinkering. Fading into the background, such a tool would provide utility without the drawbacks of existing paradigms—from waiting for translations to begin only after the completion of a sentence (i.e., offline systems that perform consecutive translations) to monotonic outputs lacking in character.

To better understand user needs when it comes to speech translation, we ground our research on the lived experiences of individuals who are dependent on translation technologies in their everyday lives. While many people use translation technologies while traveling or for other recreational purposes, this group of individuals relies on them for essential information gathering and communication. Accordingly, we interviewed 34 participants from diverse immigrant backgrounds to understand present limitations in real-world deployments of S2ST systems. The goal of this study was to understand how our interviewees, who are either Mandarin or Spanish speakers with limited English proficiency, navigate everyday communication in the United States. The narratives drawn from these interviews not only spotlight the integral role of machine translation in achieving everyday goals, but they give us an empirical window into how S2ST systems designed with naturalistic communication (i.e., with expressivity and streaming) in mind could help this population gain confidence in self-expression and spur further integration into mainstream society.

2.1.1 Meeting translation needs

As well documented by previous research, low proficiency in the languages of the receiving societies is a major source of anxiety and stress for many immigrants (Ding and Hargraves, 2009; Lueck and Wilson, 2011; Delander et al., 2005). Aside from acquiring language proficiency through learning, many tap into other strategies to bridge communication gaps (Hutchins, 2009; Orellana et al., 2003). In our interviews, we find that while most participants rely on both their personal networks and translation applications in their everyday lives, most day-to-day translation work is conducted by the latter (especially for those with higher degrees of technological literacy). Moreover, even though many commercially available translation platforms support both text and speech translation, the bulk of the translation tasks our participants perform via apps remain text-centric (i.e., translating emails, work-related documents, etc.).

This observation does not suggest that text-based translation needs supplant speech-based ones. The disparity could largely be attributed to the performance delta between text and speech-based translation tools and the lack of familiarity with speech translation functions in widely adopted translation platforms (Seamless

Communication et al., 2023). Compared to speech translation systems, text-based tools have enjoyed deeper maturity and commercial viability. User familiarity, alongside greater confidence levels in the generated outputs, drives more users to deploy text-based systems even in contexts where speech is used. It is, for example, more common for participants ($n=26/34$) to translate subtitles rather than audio speech when watching the news or television shows (even though the translation apps they use support both text and speech translation). One participant added that “speech translation just feels foreign” and that they would probably engage with it more if they saw more people using it. This is a sentiment that reverberated across the sample population.

2.1.2 Real-time translation in synchronous contexts

Despite heavy reliance on text-based translation, many participants yearn for reliable speech translation systems to help them in real time. In fact, 20 of the 34 interviewees have previously used commercially available translation platforms supporting speech. That said, using translation apps to perform consecutive speech translations was universally regarded as a workaround in time-sensitive situations, an imperfect solution to a problem. A Mandarin speaker, for example, describes a recent incident when checking out at a local grocer: “I had to give the cashier the phone, ask her to direct her question to it, and then wait for the app to translate. You can tell people around me were a little annoyed.”

For all interviewees, real-time translation would be particularly handy in social situations that require synchronous communication, whether it is face-to-face or digitally mediated. For instance, even though one could rely on text translation to render a menu legible, conversing with or responding to questions from a server poses an issue. Barriers like this not only adversely affect self-esteem but also prevent many participants from partaking in new social or cultural experiences. One participant notes that even if a speech translation system does not enable bidirectional communication, having the capability to interpret a question as it is being asked would be helpful, later adding: “at least I could gesture back or use simple English to tell someone what I want.”

For some, the lack of reliable S2ST compels them to rely on family, friends, or coworkers to help meet cross-lingual conversational needs in both informal and professional settings (Orellana et al., 2003). However, having network resources to tap into this social workaround is not a given, and even those who have access to cultural brokers (Sánchez and Orellana, 2006) express that this form of linguistic dependency stifles integration into their receiving society. In light of these constraints, one participant fantasizes that having a tool that “translates like a human, especially in circumstances where human interpreters are unavailable, could be a game-changer.”

2.1.3 Expressive translation and the preservation of vocal style & prosody

When probed on what the next generation of speech-translation technologies should look like, many participants stress that beyond simultaneous translation, future systems should enable them to communicate *naturally*. For them, *natural* communication could be interpreted in many ways—from using slang or idioms to not slowing down when directing input at a translating app. That said, the most commonly shared conceptualization ($n=29/34$) of naturalness is for S2ST systems to support prosodic preservation and the preservation of the style of one’s voice.

If text directs more attention to the content of a message, then speech more deeply emphasizes the person behind an utterance (Kraut et al., 1992). The desire for translation outputs to reflect speaker characteristics, as framed by an interviewee, suggests that S2ST systems can do way more than convey semantic information (Huang et al., 2023). Without encoding the expressive nature of speech, many participants express that a major fear of engaging with S2ST in their day-to-day lives is the risk of misaligned intent. Consider this comment by a Mandarin speaker: “Imagine if I wanted to say something sarcastically. If the system does not translate that properly, it could lead to miscommunication and misunderstandings.”

Extending this sentiment, other participants noted that faithfully reproducing vocal style and prosody in their speech breathes character into their self-expression, giving listeners a more comprehensive sense of their intent (Du et al., 2021). According to a Spanish-speaking participant, systems that go beyond *just words* can deeply transform the quality of cross-lingual communication: “Our tone is a part of our personality, and

it changes based on context and the language we are speaking. It’s also a matter of candor when we speak Spanish. We get very passionate.”

Reverberating across the interviews is the view that translation systems that deliver language coverage, expressivity, and streaming could serve as a unique tool that helps them better integrate into everyday society. Equipping those with language barriers with the ability to communicate in real-time without erasing their individuality could make prosaic activities like ordering food, communicating with a shopkeeper, or scheduling a medical appointment—all of which abilities non-immigrants take for granted—more ordinary.

2.2 Expressive and Streaming S2ST Today

Having explored the social need behind expressive and streaming S2ST systems, we now review existing efforts directed at these research areas.

2.2.1 Expressive systems

Expressive speech systems have long been of technical interest to researchers in a multidisciplinary context. Combining linguistics insights and computational methods, developing systems that can accurately produce humanlike utterances both at the semantic and paralinguistics levels becomes ever more pressing as the volume of auditory content (i.e., podcasts, audiobooks, short-form videos, etc.) and voice-assisted technologies (e.g., smart home systems, autonomous driving voice controls, etc.) are on the rise. As a technical foundation, expressive speech systems could meaningfully augment the performance of a wide variety of technologies, ranging from robotics to digital assistants.

In the translation context, expressive speech preservation with conventional cascaded S2ST systems can be realized in several ways. To preserve pre-defined word or token-level paralinguistic characteristics such as emphasis, automatic speech recognition systems (ASR) need to transcribe speech not only into text but also into pre-defined prosody labels. Subsequently, a machine translation model then translates or maps these prosody labels from the source to the target text. Finally, a text-to-speech synthesis (TTS) model synthesizes the speech output with the corresponding labels (Aguero et al., 2006; Do et al., 2017). For this pipeline to work, parallel data with aligned prosody labels is necessary.

To achieve sentence-level preservation of the style of one’s voice, TTS systems supporting cross-lingual transfer through a set of embeddings that disentangle speech nuances such as semantics (i.e., characters or phonemes), stress or tone, vocal styles, and language are typically required (Liu and Mak, 2019; Casanova et al., 2022). Recent advances in TTS have enabled voice style transfer through prompting via speech language model (Wang et al., 2023a), flow matching (Le et al., 2023), and diffusion model (Shen et al., 2023). Notably, TTS models can now be trained on non-parallel multilingual datasets and achieve cross-lingual transfer when stacked with translation models that predict semantic units (Borsos et al., 2023; Rubenstein et al., 2023a; Dong et al., 2023; Wang et al., 2023c). Relatedly, voice-aligned speech could be generated with controllable TTS models, and such data enables the training of direct S2ST systems that support translations from source speech into target speech with a consistent vocal style (Jia et al., 2022a).

Despite the recent advancements in TTS and direct S2ST (Zhang et al., 2023b; Rubenstein et al., 2023a), a comprehensive S2ST system capturing semantic translation, rhythm, pauses, and sentence-level preservation of the style of one’s voice have yet to be realized. Our work explicitly tackles preserving all such features in S2ST under a unified framework. To build our model, we first focused on addressing S2ST data paucity with aligned prosodic patterns and systematic evaluation methods. Signal-based objective metrics, such as mel-cepstral distortion (MCD), exist for TTS systems, but parallel S2ST data with aligned prosody and voice style are hard to come by (Neubig et al., 2014; Jia et al., 2022b; Ward et al., 2023). To rectify this, we devised data and textless vocal style conversion strategies to build parallel S2ST data with aligned expressivity and reference-free cross-lingual automatic evaluation methods that focus on the prosodic aspects of speech.

2.2.2 Streaming systems

In contrast to offline systems, which only start translating after the completion of a sentence, streaming systems translate as source utterances are being produced (Cho and Esipova, 2016). The biggest technical

challenge of effective streaming is striking a balance between low latency and translation quality. More specifically, a system with very low latency may miss important information, rendering a translation subpar, while a system with high latency creates excessive delays, compromising the flow of a conversation. Typically empowered by simultaneous translation policies, advanced streaming S2ST systems should dynamically decide whether to translate the next token or pause translating to absorb additional contextual information.

Research into simultaneous translation policies may be categorized into two principal categories: rule-based policies (Cho and Esipova, 2016; Dalvi et al., 2018; Ma et al., 2019a) and learnable policies. The main difference between the two policies lies in how a system waits for more input before translating. Rule-based policies rely on heuristics, such as waiting for k tokens to be read before translating, while learnable policies use algorithms such as reinforcement learning (Gu et al., 2017b) or monotonic attention to make this decision. Among the latter, monotonic-attention based models have been deemed to produce state-of-the-art performance in navigating the latency-quality trade-off (Raffel et al., 2017; Chiu* and Raffel*, 2018; Arivazhagan et al., 2019; Ma et al., 2020c). Recently, there has been a growing interest in adapting simultaneous policies to model speech inputs (Ren et al., 2020; Ma et al., 2020b, 2021; Wang et al., 2020b). To direct further attention to this underexplored area of research, recent shared tasks, such as one focused on simultaneous translation organized by the International Workshop on Spoken Language Technologies, have been established (Agarwal et al., 2023; Anastasopoulos et al., 2022, 2021; Ansari et al., 2020). These shared tasks serve as crucial avenues spurring researchers toward developing state-of-the-art models under standardized conditions.

Despite ongoing efforts dedicated to research on simultaneous translation, certain gaps require further exploration. For one, most research on streaming has focused on speech-to-text rather than speech-to-speech applications. The difference in output modality presents a technical challenge due to data and modeling constraints. Relatedly, most existing streaming models are designed in an ad hoc manner that makes them particularly sensitive to the dynamics of the offline models they are initialized on. For example, if improvements are made to a foundational offline model, it is typically quite challenging to adapt a newer streaming model to take advantage of these technical gains.

Contemporary streaming models predominantly focus on bilingual translations. However, many low-latency application scenarios consist of multiple speakers from diverse language backgrounds, calling for models that can process multilingual inputs and outputs simultaneously in an efficient manner. The development of multilingual streaming models, also an underexplored area of research, has an added advantage—cross-lingual transfer, which allows related languages to learn from one another (NLLB Team et al., 2022; Nguyen and Chiang, 2017).

Moreover, in the domain of streaming S2ST, the research has predominantly focused on a cascaded approach involving a sequential series of processing steps. However, this approach is suboptimal for real-time streaming applications, a limitation that could be alleviated by direct S2ST models (especially when the scale of training increases). Moreover, the cascaded model has issues such as compounding errors, additional disk storage, and computation time (Bentivogli et al., 2021; Seamless Communication et al., 2023). To address these issues, we combine SEAMLESSM4T v2, our multilingual and multimodal foundational model, and Efficient Monotonic Multihead Attention (EMMA), our simultaneous policy, to build a streaming translation model that performs direct translations from speech into both speech and text for many-to-many directions in real time.

2.2.3 The overarching goals of this effort

In light of the gaps delineated above, our work seeks to advance speech translation in the following ways:

1. Developing key data sets and foundational models necessary to create a unified system that enables end-to-end, multilingual, and real-time speech translation that captures a broader range of vocal style and expressive preservation.
2. Expanding language coverage both in terms of the number of supported languages and translation directions when it comes to SEAMLESSSTREAMING and SEAMLESSEXPRESSIVE translation systems (i.e., going beyond translations into English by including translation from English).
3. Maintaining systematic evaluations of our systems throughout our workflow to ensure high-quality and safe performance. This allows us to understand how to direct our efforts to make both current and

future iterations of our work more equitable and fair for different user populations.

2.3 Overview of Model Capabilities & Languages

Today, broadly accessible speech translation models cover anywhere between 21 to 113 source languages depending on the wide range of tasks involved (Zhang et al., 2023a; Rubenstein et al., 2023b). To build a unified, multimodal, and multitask model that can handle both speech and text, SEAMLESSM4T v2 covers 100 languages as speech input and 96 languages as text input. It can output 96 languages as text and 36 languages as speech. SEAMLESSEXPRESSIVE, capable of preserving rhythm, pauses, and sentence-level style of one’s voice, is equipped to handle six languages—English, French, German, Italian, Mandarin, and Spanish. As for SEAMLESSSTREAMING, our low-latency model can handle the same language coverage as SEAMLESSM4T v2 on ASR, S2TT, and S2ST tasks. We summarize information on our models’ supported capabilities and languages in Table 1. Further details on the table header are provided below.

Code. We represent each language with a three-letter ISO 639-3 code.

Language. There may be multiple ways to refer to the same language; due to formatting limitations, only one version is included below. The language names have been cross-referenced with major linguistic information platforms such as Ethnologue (Lewis, 2009) and Glottolog (Hammarström et al., 2022).

Script. We provide script information in ISO 15924 codes for writing systems.

Resource level. We categorize the speech resource level as high, medium, or low depending on the volume of available primary data for S2TT into English (with x the amount of primary data in hours, *high* if $x > 1000$, *medium* if $x \in [500, 1000]$ and *low* if $x \in [0, 500]$).

Primary data. Primary data is defined as open-source S2TT and pseudo-labeled ASR data. Absent such data, we report the language as zero-shot (when evaluating S2TT into English).

Source. We indicate whether a source language is in the speech (Sp) or text (Tx) modality, or both.

Target. We indicate whether a target language is in the speech (Sp) or text (Tx) modality, or both.

Code	Language name	Script	Resource	M4T v2		Streaming / Seamless		Expressive	
				Source	Target	Source	Target	Source	Target
afr	Afrikaans	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
amh	Amharic	Ethi	low	Sp, Tx	Tx	Sp	Tx	–	–
arb	Modern Standard Arabic	Arab	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
ary		Arab	low	Sp, Tx	Tx	Sp	Tx	–	–
arz	Egyptian Arabic	Arab	low	Sp, Tx	Tx	Sp	Tx	–	–
asm	Assamese	Beng	low	Sp, Tx	Tx	Sp	Tx	–	–
ast	Asturian	Latn	zero-shot	Sp	–	Sp	–	–	–
azj	North Azerbaijani	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
bel	Belarusian	Cyrl	high	Sp, Tx	Tx	Sp	Tx	–	–
ben	Bengali	Beng	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
bos	Bosnian	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
bul	Bulgarian	Cyrl	low	Sp, Tx	Tx	Sp	Tx	–	–
cat	Catalan	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
ceb	Cebuano	Latn	zero-shot	Sp, Tx	Tx	Sp	Tx	–	–
ces	Czech	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
ckb	Central Kurdish	Arab	low	Sp, Tx	Tx	Sp	Tx	–	–
cmn	Mandarin Chinese	Hans, Hant	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	Sp	Sp, Tx
cym		Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
dan	Danish	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
deu	German	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	Sp	Sp, Tx
ell	Greek	Grek	medium	Sp, Tx	Tx	Sp	Tx	–	–
eng	English	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	Sp	Sp, Tx
est	Estonian	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
eus	Basque	Latn	medium	Sp, Tx	Tx	Sp	Tx	–	–
fin	Finnish	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
fra	French	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	Sp	Sp, Tx
gaz	West Central Oromo	Latn	zero-shot	Sp, Tx	Tx	Sp	Tx	–	–
gle	Irish	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
glg	Galician	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
guj	Gujarati	Gujr	low	Sp, Tx	Tx	Sp	Tx	–	–
heb	Hebrew	Hebr	low	Sp, Tx	Tx	Sp	Tx	–	–
hin	Hindi	Deva	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
hrv	Croatian	Latn	medium	Sp, Tx	Tx	Sp	Tx	–	–
hun	Hungarian	Latn	medium	Sp, Tx	Tx	Sp	Tx	–	–
hye	Armenian	Armn	low	Sp, Tx	Tx	Sp	Tx	–	–
ibo	Igbo	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
ind	Indonesian	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
isl	Icelandic	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
ita	Italian	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	Sp	Sp, Tx
jav	Javanese	Latn	medium	Sp, Tx	Tx	Sp	Tx	–	–
jpn	Japanese	Jpan	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
kam	Kamba	Latn	zero-shot	Sp	–	Sp	–	–	–
kan	Kannada	Knda	low	Sp, Tx	Tx	Sp	Tx	–	–
kat	Georgian	Geor	low	Sp, Tx	Tx	Sp	Tx	–	–
kaz	Kazakh	Cyrl	medium	Sp, Tx	Tx	Sp	Tx	–	–
kea	Kabuverdianu	Latn	zero-shot	Sp	–	Sp	–	–	–
khk	Halh Mongolian	Cyrl	low	Sp, Tx	Tx	Sp	Tx	–	–
khm	Khmer	Khmr	low	Sp, Tx	Tx	Sp	Tx	–	–
kir	Kyrgyz	Cyrl	low	Sp, Tx	Tx	Sp	Tx	–	–
kor	Korean	Kore	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
lao	Lao	Lao	low	Sp, Tx	Tx	Sp	Tx	–	–
lit	Lithuanian	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
ltz	Luxembourgish	Latn	zero-shot	Sp	–	Sp	–	–	–
lug	Ganda	Latn	medium	Sp, Tx	Tx	Sp	Tx	–	–
luo	Luo	Latn	zero-shot	Sp, Tx	Tx	Sp	Tx	–	–
lvs	Standard Latvian	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
mai	Maithili	Deva	zero-shot	Sp, Tx	Tx	Sp	Tx	–	–
mal	Malayalam	Mlym	low	Sp, Tx	Tx	Sp	Tx	–	–
mar	Marathi	Deva	low	Sp, Tx	Tx	Sp	Tx	–	–
mkd	Macedonian	Cyrl	low	Sp, Tx	Tx	Sp	Tx	–	–
mlt	Maltese	Latn	low	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
mni	Meitei	Beng	zero-shot	Sp, Tx	Tx	Sp	Tx	–	–
mya	Burmese	Mymr	low	Sp, Tx	Tx	Sp	Tx	–	–

Code	Language name	Script	Resource	M4T v2		Streaming/Seamless		Expressive	
				Source	Target	Source	Target	Source	Target
nld	Dutch	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
nno	Norwegian Nynorsk	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
nob	Norwegian Bokmål	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
npi	Nepali	Deva	low	Sp, Tx	Tx	Sp	Tx	–	–
nya	Nyanja	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
oci	Occitan	Latn	zero-shot	Sp	–	Sp	–	–	–
ory	Odia	Orya	low	Sp, Tx	Tx	Sp	Tx	–	–
pan	Punjabi	Guru	low	Sp, Tx	Tx	Sp	Tx	–	–
pbt	Southern Pashto	Arab	medium	Sp, Tx	Tx	Sp	Tx	–	–
pes	Western Persian	Arab	low	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
pol	Polish	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
por	Portuguese	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
ron	Romanian	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
rus	Russian	Cyrl	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
slk	Slovak	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
slv	Slovenian	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
sna	Shona	Latn	zero-shot	Sp, Tx	Tx	Sp	Tx	–	–
snd	Sindhi	Arab	zero-shot	Sp, Tx	Tx	Sp	Tx	–	–
som	Somali	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
spa	Spanish	Latn	high	Sp, Tx	Sp, Tx	Sp	Sp, Tx	Sp	Sp, Tx
srp	Serbian	Cyrl	low	Sp, Tx	Tx	Sp	Tx	–	–
swe	Swedish	Latn	low	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
swh	Swahili	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
tam	Tamil	Taml	medium	Sp, Tx	Tx	Sp	Tx	–	–
tel	Telugu	Telu	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
tgk	Tajik	Cyrl	low	Sp, Tx	Tx	Sp	Tx	–	–
tgl	Tagalog	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
tha	Thai	Thai	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
tur	Turkish	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
ukr	Ukrainian	Cyrl	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
urd	Urdu	Arab	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
uzn	Northern Uzbek	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
vie	Vietnamese	Latn	medium	Sp, Tx	Sp, Tx	Sp	Sp, Tx	–	–
xho	Xhosa	Latn	zero-shot	Sp	–	Sp	–	–	–
yor	Yoruba	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–
yue	Cantonese	Hant	low	Sp, Tx	Tx	Sp	Tx	–	–
zlm	Colloquial Malay	Latn	low	Sp	–	Sp	–	–	–
zsm	Standard Malay	Latn	low	Tx	Tx	Sp	Tx	–	–
zul	Zulu	Latn	low	Sp, Tx	Tx	Sp	Tx	–	–

Table 1 - Seamless languages. We display the language code, name, and script, as well as the speech resource level and whether the language is supported as a source or a target in the speech and/or text modalities. Zero-shot here refers to S2TT or S2ST tasks with the language in question as source.

	Task Language Coverage				
	S2TT	S2ST	ASR	T2TT	T2ST [†]
Support	101-96	101-36	96	96-96	96-36

Table 2 - Coverage of the SeamlessM4T models. A list of supported tasks and their coverage expressed as $n_s - n_t$ where n_s and n_t are the number of languages supported as source or target respectively. [†]: the task of T2ST is evaluated zero-shot.

3. SeamlessM4T v2

The first step towards a unified SEAMLESS model, capable of expressive cross-lingual translation in real-time, starts with improving SEAMLESSM4T to give rise to SEAMLESSM4T v2—a foundational model with state-of-the-art semantic accuracy, wide language coverage, and multitasking capabilities (from and into text or speech). In terms of coverage, SEAMLESSM4T v2 supports the same tasks as SEAMLESSM4T with the same set of languages detailed in Table 1 and summarized in Table 2.

When designing the newer version of SEAMLESSM4T, adaptability to simultaneous translation was central. Due to the length mismatch between discrete acoustic units and text semantic tokens, the T2U model of SEAMLESSM4T responsible for generating units tends to hallucinate or truncate the output. This is particularly problematic if the model is only fed partial input and is tasked with generating partial outputs for real-time applications. For this reason, we pivoted in SEAMLESSM4T v2 to non-autoregressive text-to-unit decoding in order to decouple generation from output length prediction. With this non-autoregressive T2U decoder, SEAMLESSM4T v2’s S2ST inference speed has improved by 3x (see Appendix I.1) laying the ground for effective real-time translation with SEAMLESSSTREAMING.

We followed the same recipe from SEAMLESSM4T and relied on pre-training multiple blocks before finetuning them jointly as a unified model. Our unified model, previously a multitask-UNITY architecture, was upgraded to multitask-UNITY2, boasting a stronger non-autoregressive T2U model. Compared to its predecessor, UNITY2 delivers stronger T2U performance thanks to its hierarchical upsampling from subwords to characters and then to units. This upsampling makes pre-training multilingual T2U models much more data-efficient. SEAMLESSM4T v2 also used 4.5M hours of unlabeled audio data to learn its self-supervised input speech presentation with w2v-BERT 2.0 (4.5x the amount used in v1). SEAMLESSALIGN was further extended to cover more low-resource languages, enabling increased representation of these languages, ultimately improving the downstream semantic accuracy.

The key ingredients of the SEAMLESSM4T v2 recipe are:

- (a) Unlabeled, human-labeled, pseudo-labeled, or automatically aligned data used in the different pre-training and finetuning stages (Section 3.1). Figure 2 gives a bird’s eye view of the different sources of data and how they were used.
- (b) T2TT model pre-trained on NLLB data (NLLB Team et al., 2022) in nearly 100 languages (Seamless Communication et al., 2023).
- (c) Conformer speech encoder pre-trained with the w2v-BERT 2.0 algorithm. We scaled up the amount of unlabeled data from 1 million to 4.5 million hours of audio (Section 3.2.1).
- (d) X2T model trained on different sources of S2TT data (human-labeled, pseudo-labeled, and automatically aligned). This model is trained with knowledge distillation to jointly support T2TT, ASR, and S2TT by combining the models from (a) and (b) (Section 3.2.2).
- (e) UNITY2 based on a novel non-autoregressive T2U decoder architecture with hierarchical modeling of subword, character, and discrete units. UNITY2 relies on unsupervised multilingual character-to-unit alignment learning and introduces a novel span-based glancing for the T2U decoder (Section 3.3).
- (f) Multitask-UNITY2 model finetuned on speech-to-unit data (pseudo-labeled with a teacher T2U or automatically aligned) to build on the model from (c) with a student T2U model (Section 3.4).

SEAMLESSM4T-NLLB Dense transformer encoder-decoder TEXT-TO-TEXT DATA NLLB-SEED PUBLICBITEXT Automatically Aligned bitexts, MMTBT, SMTBT <i>NLLB Team et al. [2022]</i> Languages: 98 Size: 5B bitexts	w2v-BERT 2.0 Conformer UNLABELED SPEECH Publicly available data repositories Languages: 143 + Size: 4.5M hours	SEAMLESSM4T v2-T2U UNITY2's non-autoregressive T2U ASR DATA Speech audio data with transcriptions Languages: 36 Size: 34.5K hours	VOCODER HiFi-GAN unit vocoder TTS DATA Monolingual high-quality text-to-speech data Languages: 36 Size: 396 hours
X2T FINETUNING S2TT data triplets Automatically aligned S2TT pairs ASR data Size: 351K hours		S2ST FINETUNING Pseudo-labeled S2TT data Automatically aligned S2ST pairs Size: 145K hours	

Figure 2 - Data for speech translation. An overview of the pre-training and finetuning data used in SEAMLESSM4T v2.

We evaluated SEAMLESSM4T-LARGE v2 (a 2.3B-size model) across all its supported tasks [ASR, T2TT, S2TT, S2ST, T2ST (zero-shot)] and discuss its results in [Section 3.5](#).

3.1 Data for Speech Translation

In speech translation, as is the case for any other sequence modeling task, achieving state-of-the-art performances hinges on the availability of high-quality paired data used for learning. In comparison to text-to-text translation (T2TT), the amount of human-labeled speech data is scarce. To address this shortage of labeled data, we leaned on three techniques from the first version of SEAMLESSM4T ([Seamless Communication et al., 2023](#)): (1) the pre-training of different submodels on richer tasks (e.g., T2TT with SEAMLESSM4T-NLLB or unlabeled audio with w2v-BERT 2.0), (2) automatically aligning pairs, and (3) pseudo-labeling ASR data. [Figure 2](#) depicts the main building blocks of SEAMLESSM4T v2 and the different sources of data used in each pre-training or finetuning stage.

3.1.1 SeamlessAlign

We improved the SONAR speech encoders and increased their language coverage to 76 languages. This resulted in an improvement not only in the quantity of data in SEAMLESSALIGN, but also its quality and representation of low-resource languages.

Extended SONAR encoders. The backbone of speech-to-text and speech-to-speech automatic alignment is a fixed-size multilingual and multimodal sentence representation with the property that similar sentences are close in that embedding space, independently of the language and modality. We used the SONAR text encoder developed by [Duquenne et al. \(2023b\)](#), which was already successfully deployed in [Seamless Communication et al. \(2023\)](#). We trained a new set of SONAR speech encoders using the same teacher-student approach to increase the language coverage from 37 to 76, again using ASR data only. We also revisited the training data mix to remove low-quality datasets after inspection. Evaluating the various iterations of the speech encoder directly in an end-to-end automatic alignment pipeline would require to perform this alignment and then train S2TT or S2ST translation system on the aligned data, potentially comparing different thresholds of the SONAR score. This is a very compute-intensive recipe. Instead, following [Seamless Communication et al. \(2023\)](#), we evaluated our speech encoders using the SONAR text decoder and report BLEU scores for S2TT into English as a proxy for the speech encoders’ performance when used for automatically aligning pairs.

Detailed statistics for each language are shown in [Tables 61 and 62](#) under the appendix. A summary and comparison to WHISPER-LARGE-V2¹ is given in [Table 3](#). While our speech encoders perform less effectively

¹The new version v3 of Whisper seems to perform less well on S2TT

Model	deu	fra	rus	arb	isl	swh	uzn	Indian	Avg
WHISPER-LARGE-v2	34.6	32.3	27.8	25.5	9.1	7.2	6.0	13.4	19.1
SONAR	32.7	31.2	26.5	28.7	17.3	22.6	17.5	17.1	22.0

Table 3 - sacreBLEU scores on FLEURS test set for S2TT. The column *Indian* gives the average performance over 13 Indian languages (asm, ben, guj, hin, kan, mal, mar, np, pan, snd, tel, tam and urd). The average performance is calculated over 73 languages which are supported by both models.

than Whisper for 23 languages (mostly high-resource languages like German, French, or Russian), they perform substantially better on low-resource languages (like Icelandic, Swahili, Uzbek, and many Indian languages). Overall, the speech encoders exhibit very competitive S2TT performance. This is even more remarkable given that we used bottle-neck fixed-size representation rather than an attention mechanism, and performed fully zero-shot S2TT (i.e., the speech encoder was not trained using translated data and the text decoder has never seen speech input).

The speech encoders for all 76 languages are made publicly available in the SONAR repository.² See [Appendix A](#) for a model card.

Automatic alignment procedure. The speech encoders were subsequently used to perform speech-to-text and speech-to-speech automatic alignment, following the same process as introduced in [Seamless Communication et al. \(2023\)](#). Starting with 3.9 million hours of diverse raw audio originating from a publicly available repository of web data, we applied a series of preprocessing steps and segmented raw audio files into sentence-level utterances through an off-the-shelf Voice Activity Detection model ([Silerio, 2021](#)). The same language identification model was subsequently used to triage segments into language buckets, and overlapping segments were formed, following the over-segmentation approach of [Duquenne et al. \(2021\)](#).

All segments were then embedded with SONAR encoders, and indexed with the FAISS library ([Johnson et al., 2019](#)). Alignments were formed by retrieving the nearest neighbors of all elements in the forward (source in target) and backward (target in source) directions, and keeping pairs with a margin score ([Artetxe and Schwenk, 2019](#)) higher than a threshold:

$$\text{score}(x, y) = \text{margin} \left(\cos(x, y), \sum_{z \in \text{NN}_k(x)} \frac{\cos(x, z)}{2k} + \sum_{v \in \text{NN}_k(y)} \frac{\cos(y, v)}{2k} \right), \quad (1)$$

where x and y are the source and target sentences, and $\text{NN}_k(x)$ denotes the k nearest neighbors of x in the other language. We set k to 16, and the use ratio $\text{margin}(a, b) = a/b$. All code for automatically aligning data is made publicly available within the STOPES library ([Andrews et al., 2022](#)).³

The amount of automatically aligned speech is given in [Tables 61 and 62](#) in the appendix (please see the last three columns). All statistics are given with respect to a margin score threshold of 1.15. This value was obtained by limited human inspection of the aligned data and was already used in [Seamless Communication et al. \(2023\)](#). Overall, this new version of SEAMLESSALIGN has doubled its language coverage (from 37 to 76 languages) and incorporated 114,800 hours of additional data:

- English speech to non-English text (S2T eng-X)—approximately 45,300 hours
- Non-English speech to English text (S2T X-eng)—approximately 60,200 hours
- Non-English speech to English speech (S2S)—approximately 9,300 hours

Adding such large amounts of automatically aligned data can be a substantial computational challenge. Therefore, SEAMLESSALIGN can be ranked and filtered with SONAR alignment scores.

²<https://github.com/facebookresearch/SONAR>

³<https://github.com/facebookresearch/stopes>

3.1.2 Pseudo-labeling

Pseudo-labeling for S2TT. Following [Seamless Communication et al. \(2023\)](#), we circumvented the shortage of labeled S2TT data by pseudo-labeling available ASR data with a multilingual T2TT model ([Jia et al., 2019](#); [Pino et al., 2020](#)). In this case, we used NLLB-3.3B ([NLLB Team et al., 2022](#)) with the recommended decoding options. When using human-labeled data, we removed special tokens such as `<silence>` and `<no-speech>` from the verbatim transcriptions.

Pseudo-labeling for S2ST. Following [Seamless Communication et al. \(2023\)](#), we pseudo-labeled S2TT data using a text-to-unit (T2U) model. This T2U model was trained on all 36 target speech languages ([Section 3.2](#)) and can convert text into discrete units ([Tjandra et al., 2019](#); [Lee et al., 2022a,b](#); [Zhang et al., 2022](#); [Chen et al., 2023c](#)). We also used the same 10K-units vocabulary from [Seamless Communication et al. \(2023\)](#). To extract these units, features from the 35th layer of XLS-R-1B ([Babu et al., 2022](#)) are mapped to discrete categories with the k -means algorithm ($k=10,000$). The k -means centroids resemble a codebook that maps a sequence of XLS-R speech representations into a sequence of centroid indices or acoustic units. Unlike SEAMLESSM4T where we used reduced units, in SEAMLESSM4T v2 we used non-reduced (or duplicated) units (see [Section 3.3](#)).

3.1.3 Filtering

We ran the combination of human-labeled, pseudo-labeled, and automatically aligned data through a series of filters, described in detail below:

Toxicity filtering. We removed pairs with *toxicity imbalance*, i.e., when the difference in the number of toxic items detected in the source and target is above a certain threshold. For S2TT data, transcriptions were used as a proxy for speech input when counting toxic items. We set the imbalance threshold at 1.

Length filtering. We removed pairs in which the utterance is shorter than 0.1 seconds or longer than 50 seconds. We also removed pairs where the text is longer than 250 sub-words (based on the SEAMLESSM4T tokenizer).

Special characters filtering. We removed pairs in which the text contains more than 20% of emojis, more than 50% of punctuation, more than 50% of digits, or more than 50% of spaces.

Repetition filtering. We removed sentences with a contiguous repetition of a single character more than ten times. We additionally computed n -grams ($1 \leq n \leq 4$) in each text sample and filtered out the ones with less than 30% unique n -grams.

Deduplication. [Lee et al. \(2021\)](#) established that training data deduplication is critical for large language model training. In order to determine if two texts are duplicates, we applied a normalization process that removes punctuation and non-printing characters, and then replaces all digits. The filtering can remove duplicates where two data points have identical target text. This deduplication method is useful for automatically aligned data, where the same source utterances are aligned with multiple target sentences. We kept up to five pairs with duplicate targets and removed the rest.

LID filtering. We discarded pairs where the target sentences do not appear to be written in the expected languages. This can be performed automatically using a language identification model with thresholds chosen appropriately based on the reliability of LID scores for each given language. To do so, we used the LID model from [NLLB Team et al. \(2022\)](#). LID filtering was performed exclusively for Dutch, English, French, German, Italian, Polish, Portuguese, Russian, and Spanish with a confidence threshold set to 0.9.

After applying all the filters, the data used to train the SEAMLESSM4T v2 models amounts to a total of 351K hours in S2TT and 145K hours in S2ST, as described in [Table 4](#)

ASR	S2TT						S2ST			
	X-eng			eng-X			X-eng		eng-X	
	H	P	A	H	P	A	P	A	P	A
47,296	14,434	52,977	23,744	8,476	184,123	20,377	71,474	5,924	65,812	2,352

Table 4 - Total amounts of human-labeled (H), pseudo-labeled (P), and automatically aligned (A) audio data used to train the SEAMLESSM4T v2 model, measured in hours. For amounts per language, see [Tables 63](#) and [64](#).

Model	Languages	Hours	Model type	Open model
USM	over 300 [†]	12M	BEST-RQ (Chiu et al., 2022)	
MMS	1406	0.5M	wav2vec 2.0 (Baevski et al., 2020)	✓
SEAMLESSM4T-LARGE	over 143 [†]	1M	w2v-BERT 2.0	✓
SEAMLESSM4T v2	over 143 [†]	4.5M	w2v-BERT 2.0	✓

Table 5 - A comparison of multilingual speech pre-training in state-of-the-art ASR and S2TT models. [†]Estimated from the part of data that has language information.

3.2 Pre-Training

3.2.1 Self-supervised speech representation

Scaling data size for self-supervised pre-training has been empirically proven to be a relatively cheap, yet effective way to improve speech representation quality ([Zhang et al., 2023a](#)). Following such direction, we continued to add more unlabeled speech data, increasing the amount of our pre-training data from 1M hours ([Seamless Communication et al., 2023](#)) to approximately 4.5M hours.

Besides leveraging more pre-training data, we removed the random-projection quantizer (RPQ) ([Chiu et al., 2022](#)) and its associated loss previously incorporated in SEAMLESSM4T v1 ([Seamless Communication et al., 2023](#)).⁴ Akin to v1, the v2 w2v-BERT 2.0 comprises 24 Conformer layers ([Gulati et al., 2020](#)) with approximately 600M parameters and the same pre-training hyperparameters.

3.2.2 X2T: Into-text tasks

In SEAMLESSM4T, we leveraged foundational models either pre-trained on unlabeled data (w2v-BERT 2.0 for speech encoder pre-training) or trained on supervised high-resource tasks (NLLB model for T2TT) to improve the quality of transfer tasks (speech-to-text and speech-to-speech). To fuse these pre-trained components and enable meaning transfer through multiple multimodal tasks, we trained an end-to-end model with: (a) a speech encoder (w2v-BERT 2.0) postfixed with a length adapter, (b) text encoder (NLLB encoder), and (c) a text decoder (NLLB decoder). We used the same length adaptor from [Seamless Communication et al. \(2023\)](#). The text encoder was frozen, and the model was finetuned to jointly optimize the following objective functions with respect to the speech encoder parameters θ_{se} and the shared text decoder parameters θ_{td} :

$$\mathcal{L}_{S2TT}(\theta_{se}, \theta_{td}) = -\log p(y^{\text{text}} | x^{\text{text}}; \theta_{se}, \theta_{td}) = -\sum_{i=1}^{|y|} \log p(y_i^{\text{text}} | y_{<i}^{\text{text}}, x^{\text{speech}}; \theta_{se}, \theta_{td}), \quad (2)$$

$$\mathcal{L}_{T2TT}(\theta_{td}) = -\log p(y^{\text{text}} | x^{\text{text}}; \theta_{se}, \theta_{td}) = -\sum_{i=1}^{|y|} \log p(y_i^{\text{text}} | y_{<i}^{\text{text}}, x^{\text{text}}; \theta_{td}), \quad (3)$$

where x^{text} and x^{speech} are the source text and speech in the source language ℓ_s , and y^{text} is the target text in the target language ℓ_t . We additionally optimized an auxiliary objective function in the form of token-level knowledge distillation $\mathcal{L}_{KD}$ to further transfer knowledge from the strong MT model to the student speech translation task (S2TT). This loss function is defined as follows:

⁴As we scaled data from 1M to 4.5M hours, we encountered some optimization instability when RPQ was used. We decided to simply discard RPQ instead of relying on more extensive hyperparameter tuning.

$$\mathcal{L}_{\text{KD}}(\theta_{\text{se}}, \theta_{\text{td}}) = \sum_{t=1}^{|y|} D_{\text{KL}} [p(\cdot | y_{<i}^{\text{text}}, x^{\text{text}}; \theta_{\text{td}}) \| p(\cdot | y_{<i}^{\text{text}}, x^{\text{speech}}; \theta_{\text{se}}, \theta_{\text{td}})] . \quad (4)$$

Our triplets $(x^{\text{speech}}, x^{\text{text}}, y^{\text{text}})$ come mainly from pseudo-labeled ASR data (Section 3.1.2). Since we jointly trained on ASR data, handled as translation with $\ell_s = \ell_t$, we replaced the translation task for the ASR samples with auto-encoding (AE). As such, three additional losses are considered:

$$\mathcal{L}_{\text{ASR}}(\theta_{\text{se}}, \theta_{\text{td}}) = -\log p(x^{\text{text}} | x^{\text{speech}}; \theta_{\text{se}}, \theta_{\text{td}}), \quad (5)$$

$$\mathcal{L}_{\text{AE}}(\theta_{\text{td}}) = -\log p(x^{\text{text}} | x^{\text{text}}; \theta_{\text{td}}), \quad (6)$$

$$\mathcal{L}_{\text{KD-ASR}}(\theta_{\text{se}}, \theta_{\text{td}}) = \sum_{t=1}^{|y|} D_{\text{KL}} [p(\cdot | x_{<i}^{\text{text}}, x^{\text{text}}; \theta_{\text{td}}) \| p(\cdot | x_{<i}^{\text{text}}, x^{\text{speech}}; \theta_{\text{se}}, \theta_{\text{td}})] . \quad (7)$$

The final loss is a weighted sum of all six losses:

$$\mathcal{L} \propto (\mathcal{L}_{\text{S2TT}} + \mathcal{L}_{\text{T2TT}} + \mathcal{L}_{\text{KD}}) + \alpha(\mathcal{L}_{\text{ASR}} + \mathcal{L}_{\text{AE}} + \mathcal{L}_{\text{KD-ASR}}), \quad (8)$$

where α is a scalar hyperparameter dependent on the proportion of ASR data in our mix of training data.

We trained our X2T model in two stages. Stage₁ targeted training on supervised English ASR and into English S2TT data. We find that this step is necessary not only for improving the quality of X-eng translations but also eng-X translations. In fact, we hypothesized that allowing the model to focus on one target language while finetuning multilingual speech representations shields it from the interference that can propagate back from the target side. In Stage₂, we added supervised eng-X S2TT and non-English ASR data to the mix.

In SEAMLESSM4T v2, we set α (Equation (8)) to 0.04 in the first finetuning stage and 0.13 in the second stage. Our training batches present a mix of tasks (ASR or S2TT and the associated auxiliary losses), and languages (source only in the first stage and source-target in the second stage) with temperature sampling ($T = 2$). All speech encoder and text decoder parameters are finetuned for a total of 200K updates—100K in each stage.

3.3 Predicting Units with UnitY2

Similar to SEAMLESSM4T, the task of speech-to-speech translation in SEAMLESSM4T v2 is broken down into speech-to-text translation (S2TT) and then text-to-unit conversion (T2U). While UNITY relaxes the training difficulty of direct S2ST, the T2U model often hallucinates or truncates the output. This issue can be attributed to the length mismatch between the speech sequence in units and the text sequence in subwords, the former being on average 25 times longer. To reduce errors in the unit generation, we propose a new direct two-pass S2ST architecture, UNITY2, which enhances the unit generation of UNITY by a non-autoregressive (NAR) decoder that does not rely on any external aligner.

The overall architecture of UNITY2 is depicted in Figure 3. UNITY2 replaces the second-pass autoregressive unit decoder in UNITY with a NAR unit decoder. We adopted the decoder architecture of FastSpeech2 (Ren et al., 2021b) and extended it to discrete unit generation. The original FastSpeech2 decoder, designed for generating Mel-filterbank features in TTS, relies on a variance adaptor to add different variance information such as duration, pitch, and energy as conditional inputs before decoding. Given that UNITY2 is designed to model discrete units, we only added duration information with a duration predictor; other information like pitch or prosody are not well-preserved by discrete units (Polyak et al., 2021). The NAR unit decoder needs to expand the text input sequence to match the length of the unit output sequence, as such, a text-to-unit alignment is necessary. Unlike UNITY, UNITY2 predicts a duplicated (or non-reduced) unit sequence that preserves repetitive units. Although the non-reduced unit sequence is longer, fast inference with NAR unit generation can compensate for the computational overhead.

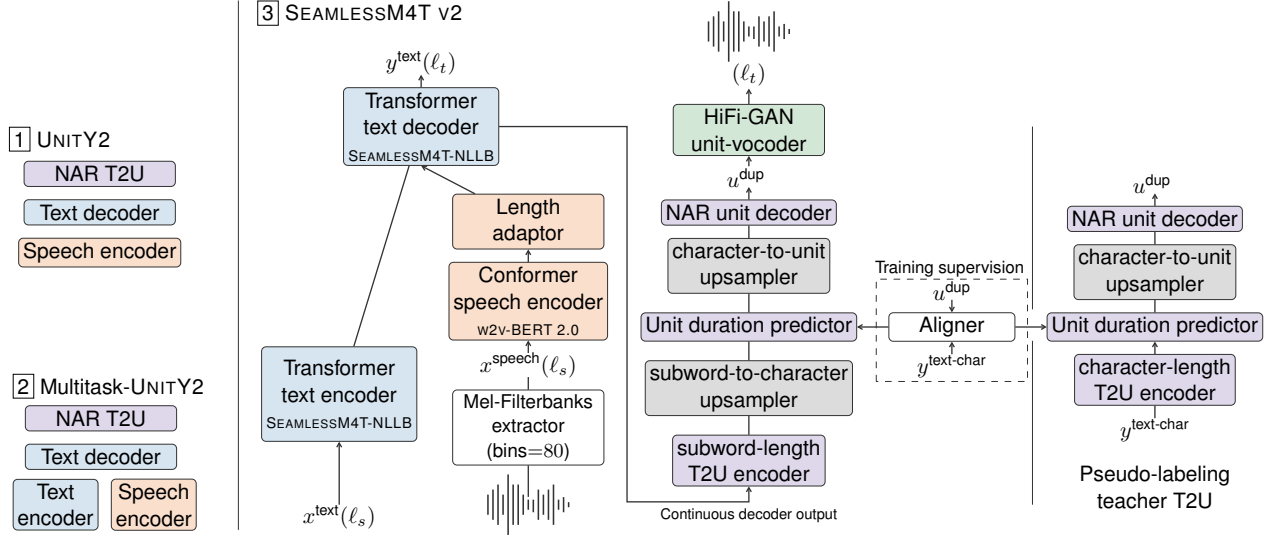


Figure 3 - Illustration of the SeamlessM4T v2 model. Panel (1) shows the three main blocks of UNITY2 with its non-autoregressive (NAR) T2U. Panel (2) shows multitask-UNITY2 with its additional text encoder. Panel (3) breaks down the components of SEAMLESSM4T v2 (a multitask-UNITY2 model) with a side panel illustration of the teacher T2U model used for pseudo-labeling.

UNITY2 starts with hierarchically upsampling of the T2U encoder output from subword-length, to character-length then to unit-length (Section 3.3.1). The *unit duration predictor*, key to the hierarchical upsampling, is supervised during training by a multilingual *aligner* based on RAD-TTS (Shih et al., 2021) (Section 3.3.2). To address the multimodality problem in the NAR generation of large-vocabulary non-reduced discrete units, we propose an efficient single-pass span-based glancing training (Section 3.3.3).

3.3.1 Hierarchical subword-to-unit upsampling

The T2U encoder in UNITY2 receives coarse subword-length representations from the X2T text decoder.⁵ As a first-pass decoder in UNITY2, its features are not suitable for describing acoustic details necessary for subsequent unit prediction. To leverage fine-grained textual information without hindering translation quality or the efficiency of the X2T decoder, we propose *hierarchical subword-to-unit upsampling*, where we upsample the subword-length T2U encoder representations to character-length then to unit-length.

Specifically, a subword-to-character upsampler **Sub2Char** repeats each subword-length representation h_i according to the number of characters in the subword y_i^{text} and adds character-level embeddings. With $y^{\text{text-char}}$, the character-length sequence corresponding to y^{text} , we compute the character-length representations h^{char} as follows:

$$d_i^{\text{char}} = f_{\text{dur}}^{\text{char}}(y_i^{\text{text}}, y^{\text{text-char}}), \quad (9)$$

$$j = \sum_{l < i} d_l^{\text{char}} + m \quad (m = 1 \dots d_i^{\text{char}}), \quad (10)$$

$$\begin{aligned} h_j^{\text{char}} &= \text{Sub2Char}(h_i, y_j^{\text{text-char}}, j) \\ &= h_i + E^{\text{char}}(y_j^{\text{text-char}}) + \frac{1}{\sqrt{d_{\text{model}}}} \cdot \text{Pos}^{\text{char}}(j), \end{aligned} \quad (11)$$

where $f_{\text{dur}}^{\text{char}}$ is a function returning the number of characters (d_i^{char}) in the i -th subword, E^{char} is a character embedding lookup table, Pos^{char} is a character-level positional embedding layer, and d_{model} is the model dimension.

⁵Subword vocabulary size of 256K in SEAMLESSM4T

Then, a character-to-unit upsampler **Char2Unit** further upsamples h^{char} to unit-length representations h^{unit} as:

$$d_j^{\text{unit}} = f_{\text{dur}}^{\text{unit}}(y_j^{\text{text-char}}, u^{\text{dup}}, A^{\text{hard}}; \theta_{\text{dur}}), \quad (12)$$

$$k = \sum_{l < j} d_l^{\text{unit}} + m \quad (m = 1, \dots, d_j^{\text{unit}}), \quad (13)$$

$$h_k^{\text{unit}} = \text{Char2Unit}(h_j^{\text{char}}, k) = h_j^{\text{char}} + \alpha \cdot \text{Pos}^{\text{unit}}(k), \quad (14)$$

where $f_{\text{dur}}^{\text{unit}}$ is a duration predictor (parameterized by θ_{dur}) that returns the number of duplicated units (d_j^{unit}) aligned with the j -th character, A^{hard} is a hard character-to-unit alignment matrix, α is a learnable scale parameter, and Pos^{unit} is a unit-level positional embedding layer. h^{unit} is used as input to the NAR unit decoder. The duration predictor $f_{\text{dur}}^{\text{unit}}$ in Equation (12) is trained to optimize $\mathcal{L}_{\text{dur}}(\theta_{\text{dur}})$, a mean square error (MSE) loss taking the duration predicted by the aligner as training target in the logarithmic domain.

3.3.2 Unsupervised multilingual character-to-unit alignment learning

For upsampling, the NAR unit decoder requires alignment (A^{hard}) between characters and units to train the unit duration predictor $f_{\text{dur}}^{\text{unit}}$. The original FastSpeech2 used a forced alignment tool (*e.g.*, Montreal Forced Aligner (McAuliffe et al., 2017)) to supervise the duration predictor. For our massively multilingual efforts, forced aligners are unavailable for many low-resource languages. To circumvent the need for external aligners, we propose an *unsupervised multilingual character-to-unit aligner*. We adapted the aligner architecture in RAD-TTS (Shih et al., 2021) to our use case. Namely, the SEAMLESSM4T v2 multilingual char-to-unit aligner is (1) modified to take discrete units and characters as inputs, (2) trained in a multilingual fashion on 35 languages, and (3) trained with curriculum learning for the alignment prior.

For a character-length sequence $y^{\text{text-char}}$ and the associated unit sequence u^{dup} , let s^{char} and s^{unit} be the outputs of the aligner’s two encoders (one for characters and one for units). A soft alignment A^{soft} is calculated as follows:

$$D_{i,j} = \|s_i^{\text{char}} - s_j^{\text{unit}}\|_2, \quad (15)$$

$$A_{i,j}^{\text{soft}} = \frac{e^{-D_{i,j}}}{\sum_k e^{-D_{k,j}}} + P_{\text{prior}}(i|j), \quad (16)$$

where P_{prior} is the Beta-binomial alignment prior to encourage near-diagonal paths (Shih et al., 2021). We disabled this alignment prior after 8k training steps to let the aligner learn a more accurate alignment later in the training. To extract a hard alignment A^{hard} from A^{soft} , the monotonic alignment search (MAS) algorithm (Kim et al., 2020) is applied.

To optimize the aligner parameters θ_{align} , we maximized the log-likelihood of all possible monotonic alignment paths $\mathcal{S}(y^{\text{text-char}})$, based on the forward algorithm. The forward sum loss $\mathcal{L}_{\text{fwd}}$ is formulated as:

$$\begin{aligned} \mathcal{L}_{\text{fwd}}(\theta_{\text{align}}) &= -\log P(\mathcal{S}(y^{\text{text-char}}) | u^{\text{dup}}; \theta_{\text{align}}), \\ &= -\log \sum_{a \in \mathcal{S}(y^{\text{text-char}})} \prod_{j=1}^{|u^{\text{dup}}|} P(a_j | u_j^{\text{dup}}; \theta_{\text{align}}), \end{aligned} \quad (17)$$

where the marginalization is efficiently implemented using a CTC loss. To enforce that A^{soft} matches A^{hard} , a binarization loss $\mathcal{L}_{\text{bin}} = A^{\text{hard}} \odot \log A^{\text{soft}}$ with $\odot$ the Hadamard product. This term is simply the KL divergence between the two alignments. $\mathcal{L}_{\text{bin}}$ is added after K_{bin} training steps.

3.3.3 Efficient span-based glancing training for NAR unit generation

Non-autoregressive sequence generation suffers from the *multimodality* problem.⁶ Previous works have addressed this problem by using iterative decoding (Lee et al., 2018), powerful generative models like

⁶Each token’s distribution depends only on the source sentence; this conditional independence assumption prevents a model from properly capturing the highly multimodal distribution of target translations (Gu et al., 2017a).

normalizing flows (Ma et al., 2019b), or diffusion-based models (Gong et al., 2023b; Reid et al., 2022). In this work, we used a single-step NAR decoder to maintain inference efficiency. Particularly, UNITY2’s NAR T2U decoder is a *Glancing Transformer* (GLAT) (Qian et al., 2021) that relaxes NAR token prediction by glancing at the ground-truth tokens. When the naive GLAT based on random masking is used for unit prediction, the task becomes trivial since adjacent units are locally correlated. To adapt GLAT to unit prediction, we propose an efficient *span-based GLAT* that operates on the character length before glancing at the units.

Given a unit prediction accuracy α , we sampled character positions to mask with a probability $1 - \alpha$. With $\mathcal{I}^{\text{char}}$ the set of the sampled positions to be masked in $y^{\text{text-char}}$, we obtained the corresponding unit positions $\mathcal{I}^{\text{unit}}$ following the aligner’s A^{hard} . We then replaced the decoder input in the $\mathcal{I}^{\text{unit}}$ positions with ground-truth unit embeddings. We demonstrate that the proposed span-based masking is more effective than random masking at the unit level. Furthermore, we propose an efficient training based on a single forward pass where α is estimated from the previous K_{glat} steps, instead of introducing a duplicate forward pass at each training step.

3.3.4 Training UnitY2’s NAR T2U and aligner

The second-pass NAR unit decoder and aligner are jointly trained with the following objective:

$$\begin{aligned} \mathcal{L}_{\text{nar}}(\theta_{T2U}, \theta_{\text{dur}}, \theta_{\text{align}}) = & \mathcal{L}_{\text{ce}}(\theta_{T2U}) + \mathcal{L}_{\text{dur}}(\theta_{\text{dur}}) + \mathcal{L}_{\text{interctc}}(\theta_{T2U}) \\ & + \mathcal{L}_{\text{fwd}}(\theta_{\text{align}}) + \mathcal{L}_{\text{bin}}(\theta_{\text{align}}), \end{aligned} \quad (18)$$

where $\mathcal{L}_{\text{interctc}}$ is a character-level CTC loss at an intermediate unit decoder layer, added to accelerate training convergence (Lee and Watanabe, 2021).

3.4 S2ST Training Setup.

Following S2ST and T2U modeling in SEAMLESSM4T, we trained two NAR T2U models for different purposes: a teacher T2U model used for unit pseudo-labeling (Section 3.1.2) and a student T2U model used for initializing the T2U sub-component in UNITY2 and finetuning on S2ST data. Both T2U models are based on the NAR decoder architecture (Section 3.3).

Teacher T2U pre-training. Since discrete unit sequences are much longer than subword sequences, we occasionally observed hallucination during unit pseudo-labeling with an auto-regressive model. NAR models, on the other hand, rarely hallucinate because duration modeling is decoupled from sequence generation.

The SEAMLESSM4T v2 teacher NAR T2U model takes characters as inputs and forgoes the subword-to-character upsampling; it takes ground-truth text for input as opposed to a text decoder output (Section 3.3.1). The teacher T2U consists of 12 encoder and 12 decoder layers.

Student T2U pre-training. The student NAR T2U takes subwords as inputs and consists of six encoder and six decoder layers. The decoder architecture is exactly the same as the unit decoder in UNITY2.

Finetuning multitask-UnityY2. In the third finetuning stage of SEAMLESSM4T v2, the multitask-UNITY2 model is initialized with the pre-trained X2T and the student NAR T2U models described above. The X2T model is frozen, and only weights corresponding to the T2U model are updated during this finetuning stage. The model is finetuned on a combination of pseudo-labeled and aligned X-eng and eng-X S2ST data totaling 145K hours (see Table 64).

The new NAR T2U architecture with the pre-trained alignment module between text and units led to superior performance and faster convergence. Given that all components are pre-trained on related tasks (S2TT, ASR, and T2U), the model converges after less than an epoch.

Multilingual HiFi-GAN unit vocoder. Unlike SEAMLESSM4T, which uses the multitask-UNITY architecture, SEAMLESSM4T v2 predicts duplicated (non-reduced) units. As such, we re-trained the unit-based HiFi-GAN vocoder from SEAMLESSM4T (Seamless Communication et al., 2023; Gong et al., 2023a) on ASR data to convert the duplicated units to waveform without performing duration prediction.

Model	size	S2TT FLEURS		S2ST FLEURS		S2ST CVSS
		(↑BLEU)		(↑ASR-BLEU)		(↑ASR-BLEU)
		X-eng (n=81)	eng-X (n=88)	X-eng (n=81)	eng-X (n=26)	X-eng (n=21)
WL-v2 (S2TT)	1.5B	17.9	–	17.8	–	29.6
WL-v3 (S2TT)	1.5B	16.9 ⁸	–			
A8B (S2TT)	8B	19.7	–			
WM (ASR) + NLLB-1.3B	2B	19.7	20.7	20.7	21.5	
WM (ASR) + NLLB-3.3B	4B	20.4	22.0	21.4	22.4	
WL-v2 (ASR) + NLLB-1.3B	2.8B	22.0	21.2	22.9	21.8	
WL-v2 (ASR) + NLLB-3.3B	4.8B	22.7	22.4	23.7	22.7	
SEAMLESSM4T-MEDIUM	1.2B	20.9	19.4	20.2	15.8	30.6
SEAMLESSM4T-LARGE	2.3B	24.1	21.8	25.8	20.9	35.7
SEAMLESSM4T v2	2.3B	26.6	22.2	29.7	26.1	39.2

Table 6 - State-of-the-art S2TT/S2ST models. Comparison against cascaded ASR +T2TT models on FLEURS S2TT, and against 2-stage and 3-stage cascaded models on FLEURS and CVSS S2ST X-eng. Results of cascaded models are highlighted in gray. We abbreviate WHISPER-LARGE as WL, WHISPER-MEDIUM as WM and AUDIOPALM-2-8B-AST as A8B.

3.5 Results and Discussion

In this section, we trained SEAMLESSM4T-LARGE v2, a 2.3B model in the multitask-UNITY2 architecture with the same coverage (i.e., tasks and languages) as SEAMLESSM4T (Seamless Communication et al., 2023). A card for this model is available in Appendix B.

We evaluated SEAMLESSM4T-LARGE v2 on all four supervised tasks (T2TT, ASR, S2TT, and S2ST), as well as the zero-shot task of text-to-speech translation (T2ST, also referred to as cross-lingual text-to-speech synthesis (Zhang et al., 2023b)).

To generate text hypotheses, we decoded with beam-search (width=5). We scored T2TT with chrF2++ and S2TT with SacreBLEU [default 13a tokenizer and character-level tokenizer for Mandarin Chinese (cmn), Japanese (jpn), Thai (tha), Lao (lao), and Burmese (mya)]. For ASR, following Radford et al. (2022), we scored normalized transcriptions and references with WER (word error rate). See metric details in Appendix H.

During S2ST and T2ST inference, we performed two-pass beam-search decoding—the best hypothesis out of the first-pass decoding is embedded with the text decoder and is sent to T2U to search for the best unit sequence hypothesis. We used a beam-width of 5 for both searches. We evaluated S2ST and T2ST accuracy with ASR-BLEU (Lee et al., 2022a) with WHISPER-LARGE as the underlying ASR model.⁷ We set the decoding temperature of Whisper at zero and used greedy decoding to ensure a deterministic behavior of the ASR model. The transcribed hypotheses, as well as the references, are normalized following (Radford et al., 2022) before computing BLEU scores (with the tokenization described for S2TT). In the following, we report averages for the per-language scores across all the evaluated tasks (see Appendix I.3).

Comparison to SeamlessM4T and cascaded models. On the set of languages supported by both SEAMLESSM4T/SEAMLESSM4T v2 and the baselines included as a reference, we compare in Table 6 the performance of our unified and direct model to that of the first version of SEAMLESSM4T, as well as cascaded models. For S2TT, the cascaded models comprise Whisper ASR models and NLLB T2TT models. For S2ST, two options were considered for cascading: (1) 3-stage with ASR, T2TT, and TTS and (2) 2-stage with S2TT and

⁷This is different from Seamless Communication et al. (2023), where WHISPER-LARGE-v2 was used for eng-X directions and WHISPER-MEDIUM was used for X-eng directions. We re-evaluated SEAMLESSM4T models here with WHISPER-LARGE for a direct comparison.

⁸We evaluated WHISPER-LARGE-v3 on S2TT FLEURS X-eng using <https://github.com/openai/whisper/>. For WHISPER-LARGE-v2, we used the results from Radford et al. (2022).

Model	CoVoST2 ($\uparrow$ BLEU)		FLORES ($\uparrow$ chrF)		
	X-eng	eng-X	X-eng	eng-X	
	($n=21$)	($n=15$)	($n=95$)	($n=95$)	
WHISPER-LARGE-v2	29.1	x	NLLB-1.3B	59.3	48.2
AUDIOPALM-2-8B-AST	37.8	x	NLLB-3.3B	60.6	49.6
SEAMLESSM4T-MEDIUM	29.8	26.6	SEAMLESSM4T-MEDIUM	55.4	48.4
SEAMLESSM4T-LARGE	34.1	30.6	SEAMLESSM4T-LARGE	60.8	50.9
SEAMLESSM4T-LARGE v2	36.6	31.7	SEAMLESSM4T-LARGE v2	59.2	49.3

ASR ($\downarrow$ WER)				
	FLEURS-77 ($n=77$)	FLEURS-60 ($n=60$)	FLEURS-54 ($n=54$)	FLEURS-41 ($n=41$)
WHISPER-LARGE-v2	41.7	24.0	43.7	25.0
WHISPER-LARGE-v3	34.9	17.2	35.6	17.0
MMS-L1107-CCLM-LSAH	–	–	18.7	16.5
SEAMLESSM4T-MEDIUM	21.9	16.4	22.0	16.4
SEAMLESSM4T-LARGE	22.6	16.6	23.2	16.9
SEAMLESSM4T-LARGE v2	18.5	12.8	19.1	13.1

Table 7 - Multitasking X2T results. Performance of SEAMLESSM4T-LARGE on X2T tasks (S2TT, ASR and T2TT) compared to SOTA direct translation models. For MT, we average chrF scores over the supported written languages in SEAMLESSM4T ($n=96$). For FLEURS ASR, we report the average normalized WER over languages supported by both SEAMLESSM4T and Whisper Large (WL) (FLEURS-77). For MMS, we report the results of the MMS-L1107-CCLM-LSAH model (CTC-based with an n-gram language model for each language) on FLEURS-54. For a direct comparison with WHISPER-LARGE-v3, we average over whisper’s reported WER scores on FLEURS-60. To compare all ASR models on a common benchmark, we included averages over FLEURS-41.

TTS. We used YOURTTS for English-TTS (Casanova et al., 2022) and MMS’s TTS models for non-English⁹ TTS (Pratap et al., 2023).

In FLEURS, SEAMLESSM4T v2 achieves state-of-the-art performance in S2TT, improving in X-eng by 10% over SEAMLESSM4T-LARGE, and by more than 17% over the strongest cascaded model (WHISPER-LARGE-v2 + NLLB-3.3B). When compared against direct models (*e.g.*, Whisper and AudioPaLM), SEAMLESSM4T v2 significantly outperformed both in X-eng directions by more than 35%.¹⁰

In speech-to-speech translation, SEAMLESSM4T v2 improves over SEAMLESSM4T-LARGE in FLEURS by more than 15% in X-eng and 25% in eng-X. Compared to the strongest cascaded models, this is an improvement of 25% and 15% in X-eng and eng-X, respectively. Results on CVSS show a similar trend and a consistently strong performance with generalizability to other domains.

Multitasking results. We compare in Table 7 the performance of SEAMLESSM4T v2 to that of state-of-the-art models in T2TT and ASR tasks. Evaluated for FLEURS ASR, on the overlapping 77 languages between WHISPER-LARGE-v2 and SEAMLESSM4T, SEAMLESSM4T-LARGE v2 improved over SEAMLESSM4T-LARGE by a relative -21% WER and over WHISPER-LARGE-v2 by a relative -56% WER. For comparison against MMS, we also report the average on FLEURS-54, where SEAMLESSM4T-LARGE v2 improves over SEAMLESSM4T-LARGE by a relative -19% WER, closing the gap with MMS’s best model (MMS-L61-noLM-LSAH) to -0.4 WER. We also compared SEAMLESSM4T-LARGE v2’s ASR performance to the recently released WHISPER-LARGE-v3. Evaluated on 60 languages from FLEURS (as reported in the release¹¹), SEAMLESSM4T-LARGE

⁹Only 26 of our 35 supported languages are serviced by MMS’s TTS models

¹⁰We evaluated the recently released WHISPER-LARGE-v3 on FLEURS’s S2TT and found it to be worse than WHISPER-LARGE-v2.

¹¹<https://github.com/openai/whisper/discussions/1762>

Model	FLEURS T2ST ($\uparrow$ ASR-BLEU)	
	X-eng ($n=88$)	eng-X ($n=26$)
NLLB-1.3B	35.0	22.7
NLLB-3.3B	36.4	23.7
SEAMLESSM4T-MEDIUM	26.3	18.4
SEAMLESSM4T-LARGE	34.1	21.8
SEAMLESSM4T-LARGE v2	35.9	27.6

Table 8 - Zero-shot Fleurs T2ST. We report the average ASR-BLEU of SEAMLESSM4T-LARGE on FLEURS T2ST.

Resource-level	S2TT X-eng $\uparrow \Delta$ BLEU	S2ST X-eng $\uparrow \Delta$ ASR-BLEU	ASR $\downarrow \Delta$ WER
Low ($n=42$)	+2.8	+4.3	-7.5
Medium ($n=26$)	+3.0	+4.5	-4.7
High ($n=16$)	+1.7	+2.9	-2.9

Table 9 - Improvement from SeamlessM4T to SeamlessM4T v2. Delta of performance in FLEURS’s S2TT X-eng, S2ST X-eng and ASR between SEAMLESSM4T-LARGE and SEAMLESSM4T-LARGE v2.

v2 improved over WHISPER-LARGE-v3 by -4.4% WER.

Evaluated for T2TT, SEAMLESSM4T-LARGE v2’s performance on FLORES drops by -1.6 chrF2++ in both X-eng and eng-X when compared to SEAMLESSM4T-LARGE. Its T2TT accuracy is still, however, on par with the equally-sized NLLB-1.3B for X-eng and NLLB-3.3B for eng-X.

Evaluated on CoVoST2 (Wang et al., 2021), a multilingual S2TT benchmark dataset, SEAMLESSM4T-LARGE v2 improved over SEAMLESSM4T-LARGE by +2.5 BLEU in X-eng directions and by +1.1 in eng-X directions. In X-eng directions SEAMLESSM4T still lags behind AUDIOPALM-2-8B-AST (-1.2 BLEU).

Zero-shot text-to-speech translation. We next evaluated SEAMLESSM4T-LARGE v2 on the task of T2ST in a zero-shot way. Given that FLEURS collected three recordings by three different native speakers for each sample, we randomly select one for the task of T2ST (the input being text). We report in Table 8 a comparison between SEAMLESSM4T models and cascaded models with NLLB and either YOURTTS (English TTS) or MMS (non-English TTS) for synthesizing translated text. We averaged ASR-BLEU scores over 88 X-eng directions (the overlap between FLEURS and the languages supported by SEAMLESSM4T v2). We also averaged ASR-BLEU over 26 eng-X directions (the overlap between our 35 and the languages supported by MMS’s TTS models). SEAMLESSM4T-LARGE v2 improved by a large margin over SEAMLESSM4T-LARGE (+1.8 and +5.8 ASR-BLEU points in X-eng and eng-X respectively). Compared to cascaded models, SEAMLESSM4T-LARGE v2’s zero-shot capability is on par with NLLB-3.3B + YOURTTS in X-eng, and outperforms NLLB-3.3B + MMS by more than +3.9 ASR-BLEU points in eng-X.

Results by resource-level. We show in Table 9 the improvements in FLEURS S2TT X-eng, S2ST X-eng and ASR achieved in SEAMLESSM4T-LARGE v2 when buttressed with additional supervised data (mostly automatically aligned) and unlabeled data used to train our w2v-BERT 2.0 speech encoder. Our efforts to increase supervised and self-supervised data targeted low- and medium-resource languages. Overall, SEAMLESSM4T-LARGE v2 improved on low-resource languages by an average of 2.8 BLEU points, 4.3 ASR-BLEU points and -7.5 WER in the three tasks respectively. As for medium-resource languages, it improved by an average of 3.0 BLEU points, 4.5 ASR-BLEU points and -4.7 WER respectively.

Ablation on the input representations for T2U. We investigated better input and output representations for both AR and NAR T2U models. To do so, we compared subword and character as input with reduced and non-reduced units as output in Table 10a. We found that the previous setting with subword input and reduced

unit was the best for the AR T2U model, while character input and non-reduced unit were the best for the NAR T2U model. The best NAR T2U model outperformed the best AR T2U model by 35% in ASR-WER.

Model	Text input tokenization	Output units deduplication	↓ASR-WER ($n=32$)
AR T2U	Subword	Reduced	20.79
	Subword	Non-reduced	24.78
	Character	Reduced	35.49
	Character	Non-reduced	78.35
NAR T2U	Subword	Reduced	16.66
	Subword	Non-reduced	16.54
	Character	Reduced	13.91
	Character	Non-reduced	13.41

(a) A comparison of input and output representations in teacher T2U modeling.

Model	↓ASR-WER ($n=32$)
NAR T2U	13.41
w/o GLAT	14.97
w/o Span-based masking	15.17
w/o Efficient GLAT	13.54
w/o InterCTC	13.92

(b) Ablation studies in character-level teacher NAR T2U modeling.

Table 10 - Ablation studies in T2U modeling. In each set of experiments, we calculated ASR-WER in 32 of 36 languages since ASR performs poorly (*i.e.*, WER > 50%) for Bengali (ben), Maltese (mlt), Telugu(tel) and Northern Uzbek (uzn).

Ablation on the modeling of NAR T2U. We next conducted an ablation study of the proposed NAR T2U modeling in Table 10b. We confirmed that GLAT significantly improved intelligibility and both span-based masking and character-level InterCTC also contributed to further improvement. Efficient GLAT did not degrade ASR-WER despite a single forward pass.

UnitY2’s multilingual char-to-unit aligner. The UNITY2-based aligner component, used as a duration teacher in the T2U training, presents itself as a universal tool to align arbitrary text-audio pairs for any downstream task. The presence of extremely large, unlabeled audio corpora makes this tool very attractive for pseudo-labeling. We release a multilingual aligner component that supports all 36 target languages of SEAMLESSM4T v2, together with a front end for alignment extraction. The front end uses a character-based Sentence-piece model to tokenize a raw text sequence and a 10K acoustic unit extractor, which outputs a discrete unit sequence from SEAMLESSM4T v2’s unit space. We found that our aligner also works pretty well when using a normalized text. A model card describing the aligner component can be found in Appendix F. Figure 4 shows an example of a Russian audio sample aligned with its transcription, where the waveform exhibits variable speech rate. In this work, we utilized this alignment extraction tool as the core component behind the automatic pause alignment evaluation (see Section 7.1 for more details).

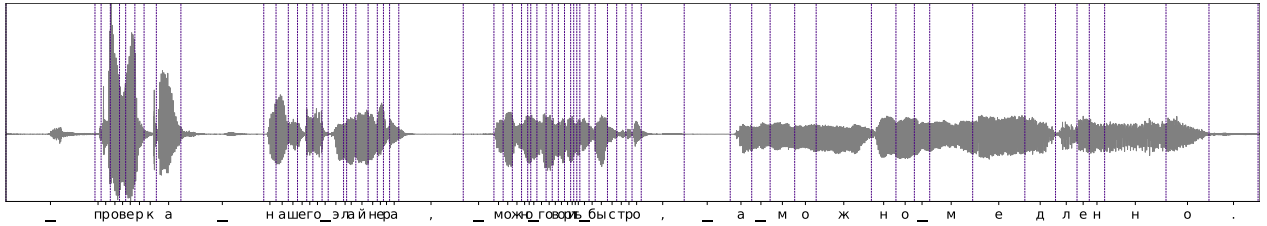


Figure 4 - Visualization of an alignment with UnitY2’s aligner. Example of a Russian audio aligned with its transcription “проверка нашего элайнера, можно говорить быстро, а можно медленно.” The purple vertical lines show the predicted character boundaries.